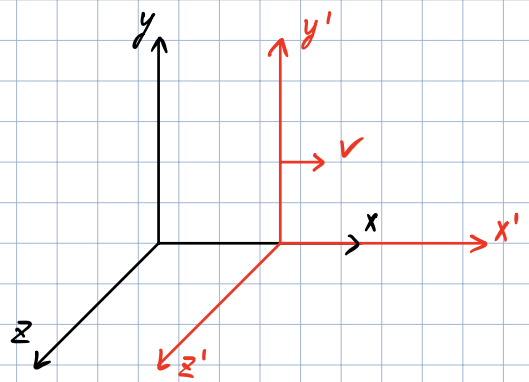


Lorentz transformation :

$$\begin{cases} x' = \gamma(x - vt) \\ y' = y \\ z' = z \\ t' = \gamma\left(t - \frac{v}{c^2}x\right) \end{cases} \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

inverse transformation :

$$\begin{cases} x = \gamma(x' + vt') \\ y = y' \\ z = z' \\ t = \gamma\left(t' + \frac{v}{c^2}x'\right) \end{cases}$$

The relativity of simultaneity

Two events occur same time in S,

$$t_A' = t_B' + \frac{\gamma v}{c^2}(x_B - x_A).$$